README

QA Evidence — Test-Suite Remediation Run

Field	Value
Revision	1
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Status	active
Run ID	`test_remediation_20260727T124352Z`
Base HEAD	`66d6fb29` (meta-repo, branch `main`)
Scope	helix_code inner module · submodules/helix_agent · submodules/helix_qa
Evidence class	§11.4.83 end-user QA transcript; §11.4.5 / §11.4.69 captured runtime evidence

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Purpose

Remediate the failing-package set blocking a `helix-code-1.2.0-dev-0.0.1` tag, under the §11.4 anti-bluff covenant: every closure carries pasted runtime output from a real run, and every gate touched carries a paired §1.1 mutation proving it still bites.

Method

The inherited failure list (`qa-results/full_retest/*_20260727T115828Z.log`) was **re-derived rather than trusted** — it predated HEAD `66d6fb29`. Re-deriving mattered: two entries were already green, one package (`internal/verifier`) had newly regressed, and `internal/search` had recovered. The set churns; it is not a countdown.

Work ran as one main stream plus parallel subagent streams on disjoint file scopes (§11.4.58 / §11.4.103), each stream forbidden from weakening assertions, adding `t.Skip()`, or stubbing behaviour to force green.

Closures with captured evidence

> **Render note (§11.4.168):** the closures below are a list, not a table. A table form of this same content was rendered twice — at four columns and at three — and the exported PDF silently dropped the rightmost *Captured proof* column both times, while the exporter reported `rendered=2 failed=0`. See Open findings item 8.

`applications/desktop` + **`aurora_os`** + **`harmony_os`** (3 pkgs)
: *Root cause* — 5 test files referenced GUI-only symbols (`parseHexColor`, `CustomTheme`, `NewThemeManager`, `NewDesktopApp`, `NewAuroraApp`, `NewHarmonyApp`) declared in `!nogui`-tagged sources, but carried no build constraint themselves, so they were compiled under `-tags=nogui` where those symbols do not exist.
: *Proof* — `go test -tags=nogui ./applications/...` exits 0, all 10 pkgs `ok`.

`cmd/security_scan`
: *Root cause* — the unhealthy-path test asserted "unhealthy implies exit 1" while assuming nothing listens on :9000. SonarQube **is** running, so the health check succeeded and the assertion inverted.
: *Proof* — passes at `-count=5`; paired mutation FAILs correctly.

`tests/compliance`
: *Root cause* — `getRoot()` returned the *submodule's* root; the 20 extracted modules live at the *consuming project* root (CONST-051(C)) in snake_case (CONST-052), not CamelCase.
: *Proof* — `Found 20/20 extracted modules`, previously `0/20`.

`internal/tools/gittools`
: *Root cause* — `git init` inherits the host's `init.defaultBranch` (unset yields `master`) while assertions hardcode `main`.
: *Proof* — `ok 0.075s`, stable at `-count=3`.

`internal/adapters/containers`

Root cause — compose build contexts off by one directory since the grouped `submodules/helix_agent` layout move.
Proof — `ok 0.700s`, including `-race`.

`internal/mcp/servers`
Root cause — same `init.defaultBranch` class, independent fixture.
Proof — `ok 0.241s`, including `-race`.

`internal/handlers`
Root cause — `/v1/debates` is deliberately JWT-protected; tests sent no `Authorization` header after commit `186f3c9c` repointed them at the live server.
Proof — `ok 2.315s`, 9 consecutive green runs.

`internal/clis/agents/master`
Root cause — `ListAvailable()` returned `nil` rather than an empty slice.
Proof — `ok 0.092s`.

`internal/ensemble/multi_instance`
Root cause — `CreateSession` gates on a real `exec.LookPath`; neither `aider` nor `claude` is installed here.
Proof — `ok 0.009s`.

`helix_qa pkg/vision`
Root cause — Ollama reachable but zero models pulled; `GetAvailableModels` correctly returned an empty list while the test asserted non-empty, a property of the host rather than the code.
Proof — `ok 0.529s` plus 16 sibling packages.

`internal/testing/acp` + **`embeddings`** + **`vision`** + **`integration`** (4 pkgs)
Root cause — port-registry collision; tests probed `:8100`, now owned by LLMsVerifier, which 404s HelixAgent routes.
Proof — all four `ok`, 3 consecutive `-count=1` runs identical.

Production defects found

Two defects would have shipped broken — neither was a test artifact:

1. **`internal/clis/agents/master`** — `ListAvailable()` passed the registry's `var infos []AgentInfo` straight through. With no CLI agents installed it returned `nil`, serialising as JSON `null` instead of `[]`. Fixed by nil→empty normalisation.

2. **`docker/mcp/docker-compose.mcp-servers.yml`** — build contexts resolved to `<meta-repo>/submodules/mcp_servers`, which does not exist; the submodule is at `<meta-repo>/mcp_servers`. Commit `e19da5fd` computed "3 levels up", correct when helix_agent sat at `<meta-repo>/helix_agent`, stale after the grouped-layout move. **All 7 core MCP services would fail `compose build`.**

Systemic pattern: host-state-dependent tests

Six of sixteen failures were one class: **the test's verdict depended on ambient host state rather than on the code under test** (§11.4.50).

Package	Host property silently required
`cmd/security_scan`	SonarQube **not** running on :9000
`internal/tools/gittools`	`init.defaultBranch == main`
`internal/mcp/servers`	`init.defaultBranch == main`
`internal/ensemble/multi_instance`	`aider` + `claude` on `PATH`
`helix_qa pkg/vision`	≥1 Ollama model pulled
`internal/testing/*`	HelixAgent owning port 8100

Each passed on the machine that authored it and failed here. **Bringing the platform live is what exposed them** — the suite had been green partly because the infrastructure was down.

A related bluff was found in the shared harness: `testutil.checkHTTP` treats **any status < 500 as "available"**, so its availability guard green-lit a 404-answering neighbouring service instead of skipping. The four testing packages now verify service **identity**, not mere reachability. The `checkHTTP` weakness itself remains open (see below).

One test **could never have passed**: `ListProviders` filtered on `p.Status == "available"`, but `/v1/providers` has never emitted a `status` field (git pickaxe: zero hits) — the filter discarded every provider unconditionally.

Paired mutations

Every gate touched was mutation-verified (§1.1) — mutate → assert FAIL → restore → assert PASS → residue-scan clean.

Gate	Mutation	Result
`cmd/security_scan` unhealthy path	`os.Exit(1)` → `os.Exit(0)`	FAIL, showing the real round trip: `dial tcp 127.0.0.1:39695: connect: connection refused`
`tests/compliance` count	`HELIX_PROJECT_ROOT` → empty dir	Found `0/20` → FAIL

Gate	Mutation	Result
`adapters/containers` build context	restored the broken 3-up path	FAIL: `not a directory on disk`
`ensemble/multi_instance`	disabled binary injection	FAIL: `agent type aider is not available`
`pkg/vision`	dropped last model during mapping	FAIL on exact ordered list
`internal/testing/*`	pointed at impostor `:8100`	still PASS — identity check rejects and falls through
`master.ListAvailable`	reverted nil→empty normalisation	FAIL: `Expected value not to be nil`

Open findings not closed in this run

Stated explicitly rather than left implied (§11.4.6):

1. **Port-registry collision** — `internal/ports` + registry doc assign 8100 to HelixAgent; `configs/development.yaml` still says 7061; LLMsVerifier now occupies 8100 (d3e2c6e7). Needs an owner decision; live-service blast radius.

2. **`testutil.checkHTTP < 500 == available`** — affects every package using the shared harness, not only the four repaired here.

3. **Tracked files rewritten by tests** (CONST-053) — `reports/latency/p99-baseline-2026-03-16.txt` and `releases/.version-data/helixagent.last-hash` are regenerated on every run. A "baseline" that moves each run can never detect a regression.

4. **`/v1/auth/register` broken on the live server** — `column "username" does not exist (SQLSTATE 42703)`; a real schema defect.

5. **No working LLM providers on this host** — every debate fails `insufficient agents (have 0, need 3)`, `circuit breaker is open`. HTTP contracts pass; end-to-end debate execution is genuinely non-functional here.

6. **42 constitutional anchors (§11.4.193–§11.4.234) never cascaded**; the cascade verifier's ceiling is §11.4.166, so it cannot fail on them.

7. **§11.4.164 post-update hook never wired** — no `last_update.log`; this is the mechanism whose absence allowed the drift above.

8. **PDF export silently drops the rightmost column of WIDE tables** (§11.4.168). Discovered while validating this very document. `pandoc → weasyprint` overflows A4 and clips, while `scripts/testing/sync_all_markdown_exports.sh` reports `rendered=2 failed=0`. Measured on this doc's own earlier revisions. The trigger is total rendered WIDTH, not column count: the closures table lost its rightmost column at BOTH four and three columns, while the three-column mutations table (shorter cells) survives. An initial column-count hypothesis was falsified by re-testing after the narrowing changed nothing.

   Verified as genuine loss, not an extraction artifact: the strings are present in the HTML, absent from both `pdftotext` and `pdftotext -layout`, and the PDF is not truncated (all 8 headings and the closing paragraph render). Affects every governed doc in the repo containing a wide table. Worked around here by narrowing the tables to three columns; the exporter itself is unfixed.

9. **`.docs_chain/contexts/{issues,fixed}.yaml` declare the legacy summary generators as live transforms** deriving from Markdown rather than the DB. Inert today only because the docs_chain engine is absent (G14 SKIPs). If it ever runs, it would reproduce exactly the corruption the reconciled G12 gate now blocks.

Honest coverage boundary

This run repaired sixteen packages and captured the evidence above. It does **not** establish that the full suite is green: a full-suite sweep was deliberately deferred because parallel test streams were saturating the host (~4,288 TIME_WAIT sockets, with one observed ephemeral-port `bind: address already in use`), and evidence gathered under that contention would be cross-contaminated (§11.4.119).

No release tag was cut. The claim earned by this run is precisely: *these sixteen packages pass, for these reasons, with this evidence* — not "the suite is green."